

Fyberbuild Surveyor Portal — User Flows for Linear Tickets

Use this document to create Linear tickets/epics per role. Each flow maps to one or more implementable tickets.

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1. Authentication & Routing

AUTH-01: Login & Role-Based Redirect

- o User enters credentials on `/login`
- o System authenticates and determines role (PM, Surveyor, Validator, Subcontractor)
- o Redirect to role-specific portal: `/admin`, `/surveyor`, `/validator`, `/subcontractor`
- o Unauthorized access to other portals blocked

AUTH-02: Session Management

- o Token-based session persistence
- o Auto-logout on token expiry
- o "Remember me" option
- o Redirect to login on expired session

AUTH-03: Role-Based Access Control (RBAC)

- o Each API call validates role permissions server-side
- o Frontend hides UI elements based on role
- o Permission matrix enforced (see appendix)

2. Surveyor Flows

Epic: Surveyor Dashboard

SVR-01: View Dashboard with KPI Cards

As a Surveyor I want to see my dashboard with KPI metrics **so that** I have an overview of my workload. **Acceptance Criteria:**

- Display KPI cards: Total Assigned, Active (in_progress + visited), Pending (to_do + appointment), In Review (validation_f49 + validation_client), Issues (rework + rejected + final_no_entry)
- KPIs update in real-time when survey statuses change
- Mobile: cards stack vertically

SVR-02: Filter Surveys by Status Category

As a Surveyor I want to filter my survey list by status category **so that** I can focus on specific work items.

Acceptance Criteria:

- Filter buttons: Active, Pending, Review, Done, Issues
- Survey card list updates immediately on filter selection
- Active filter visually highlighted
- Project dropdown filter available alongside status filters

SVR-03: View Survey Cards List

As a Surveyor I want to see a list of my assigned surveys as cards **so that** I can pick which one to work on.

Acceptance Criteria:

- Each card shows: tsg_id, full address, status badge (color-coded), building type indicator
- Cards sorted by priority, then by status
- Click card → opens survey form
- Mobile: full-width cards in scrollable list

SVR-04: Download Tomorrow's Surveys

As a Surveyor I want to download scheduled survey imports **so that** I have the latest assignments. **Acceptance Criteria:**

- "Download Tomorrow" card visible on dashboard
- Shows count of scheduled imports
- Click to fetch and sync new assignments
- Success/error toast notification

Epic: Survey Form

SVR-05: Open Survey Form

As a Surveyor I want to open a survey form **so that** I can fill in site data. **Acceptance Criteria:**

- Click survey card → full survey form opens
- Desktop: sidebar collapses, form takes full width
- Mobile: bottom nav hides, back button appears in header
- Form sections rendered based on current survey status (see section visibility rules)

SVR-06: Auto Status Transition — First Save

As a Surveyor when I save a survey for the first time (status = to_do) **then** the status auto-transitions to **on_going**.

Acceptance Criteria:

- o First field save on a **to_do** survey changes status to **on_going**
- o Status badge updates immediately
- o Dashboard KPIs reflect the change
- o Grouped MDU statuses also update

SVR-07: Autosave Survey Data

As a Surveyor I want my form data to auto-save **so that** I don't lose work. **Acceptance Criteria:**

- o Debounced autosave triggers 3 seconds after last field change
- o "Saving..." indicator visible during save
- o "Saved" confirmation after successful save
- o Save disabled for statuses: inactive, sv, na, to_do (until first explicit save)
- o Works offline (queues saves for sync)

SVR-08: Navigate Between Sections

As a Surveyor I want to move between survey form sections **so that** I can fill data section by section. **Acceptance Criteria:**

- o "Save and Continue" button advances to next section
- o "Previous" button returns to prior section
- o Section navigation sidebar/stepper shows progress
- o Cannot advance if current section has missing required fields (validation error shown)

SVR-09: Section Visibility Based on Status

As a Surveyor I want only relevant sections to be visible **so that** I'm not overwhelmed. **Acceptance Criteria:**

- o **Always visible** (any status): building_owner, photo_letterbox, bordje_syndic
- o **Hidden** in early statuses (inactive, sv, na, to_do, appointment): client_info, building_info, distribution_zone, facade_street, existing_telecom, execution_quantities, legend, photo sections (facade, floorplan, fire_dept, misc), technical sections, engineering_plans, statement_agreement
- o **Conditionally visible**: appointment_info visible in to_do, rework, visited, on_going, appointment
- o **Post-on_going**: All remaining sections become visible

SVR-10: Fill Building Owner Section (Required)

As a Surveyor I want to record building owner details **so that** we have owner contact info. **Acceptance Criteria:**

- o Owner name field (required)
- o Contact details (phone, email)
- o Always visible regardless of status
- o Validation: cannot submit survey without owner_name

SVR-11: Fill Client Info Section

As a Surveyor I want to record client information **so that** the survey has company and contact context. **Acceptance Criteria:**

- o Company name (required)
- o Contact name (required)
- o Contact phone, email
- o Visible after status = on_going

SVR-12: Fill Building Info Section

As a Surveyor I want to record building classification data **so that** engineering can determine build type. **Acceptance Criteria:**

- Number of units input
- Number of floors input
- Building type auto-calculated: 9–16 units = small, 17–48 = large, >48 = extra_large
- Building type badge displayed (read-only, auto-set)

SVR-13: Fill Distribution Zone Section

As a Surveyor I want to record MRO/POP area details. **Acceptance Criteria:**

- MRO zone selection
- POP zone selection
- Distribution type

SVR-14: Fill Facade & Existing Telecom Sections

As a Surveyor I want to assess facade conditions and existing telecom infrastructure. **Acceptance Criteria:**

- Facade street assessment fields
- Existing telecom infrastructure details
- Visible after status = on_going

SVR-15: Fill Execution Quantities Section

As a Surveyor I want to estimate cost-related quantities **so that** financials can be projected. **Acceptance Criteria:**

- Quantity input fields per material/labor type
- Auto-calculation of subtotals where applicable

SVR-16: View Legend Section (Read-Only)

As a Surveyor I want to reference the legend **so that** I use correct codes/symbols. **Acceptance Criteria:**

- Legend section is read-only
- No edit controls
- Always accessible when visible

SVR-17: Fill Technical Sections (Underground, Facade Dist, Tech Room, Cable Trajectory)

As a Surveyor I want to fill in technical infrastructure sections. **Acceptance Criteria:**

- Underground introduction details
- Facade distribution assessment
- Technical room details
- Cable trajectory data
- Each section has its own fields and validation

Epic: Photo Management

SVR-18: Upload Photos — Letterbox (Required)

As a Surveyor I want to upload letterbox photos **so that** there's visual evidence of building identification. **Acceptance Criteria:**

- Photo upload control (camera capture on mobile, file select on desktop)
- Minimum 1 photo required
- Annotation capability (draw/mark on photo)

- Remarks text field
- Always visible regardless of status

SVR-19: Upload Photos — Bordje Syndic (Required)

As a Surveyor I want to upload syndic sign/board photos. **Acceptance Criteria:**

- Same pattern as letterbox: upload + annotation + remarks
- Minimum 1 photo required
- Always visible

SVR-20: Upload Photos — Facade

As a Surveyor I want to upload facade photos **so that** the building exterior is documented. **Acceptance Criteria:**

- Multiple photo upload
- Annotation on each photo
- Remarks field
- Visible after status = on_going

SVR-21: Upload Photos — Floorplan Canvas

As a Surveyor I want to upload/draw floorplan photos **so that** the building layout is documented. **Acceptance Criteria:**

- Photo upload with canvas annotation
- Drawing tools for marking cable routes on floorplan
- Visible after status = on_going

SVR-22: Upload Photos — Fire Department

As a Surveyor I want to upload fire department compliance photos. **Acceptance Criteria:**

- Photo upload + annotation + remarks
- Visible after status = on_going

SVR-23: Upload Photos — Miscellaneous

As a Surveyor I want to upload any additional photos not covered by other sections. **Acceptance Criteria:**

- Multiple photo upload
- Label/tag each photo
- Remarks field

Epic: Visit & Appointment Management

SVR-24: Add Site Visit

As a Surveyor I want to log a site visit **so that** visit history is tracked. **Acceptance Criteria:**

- "Add Visit" action available on survey form
- Records visit timestamp automatically
- Entry/no-entry toggle
- On status = on_going, adding visit → auto-transition to `visited`
- 3+ "No entry" visits → auto-transition to `final_no_entry`

SVR-25: Delete Site Visit

As a Surveyor I want to remove an incorrectly logged visit. **Acceptance Criteria:**

- Delete button on each visit record

- Confirmation dialog before deletion
- Status does not revert on deletion

SVR-26: Add Appointment

As a Surveyor I want to schedule an appointment with the building owner/syndic. **Acceptance Criteria:**

- Date and time picker
- Auto-transition status to `appointment`
- Appointment visible on dashboard

SVR-27: Delete Appointment

As a Surveyor I want to cancel a scheduled appointment. **Acceptance Criteria:**

- Delete button with confirmation
- Status does not revert

Epic: Survey Submission & Rework

SVR-28: Submit Completed Survey

As a Surveyor I want to submit my completed survey **so that** it goes to validation. **Acceptance Criteria:**

- "Complete Survey" button visible on final section
- Pre-submission validation: all required fields + photos present
- If validation fails: show list of missing items, block submission
- On success: status → `visited` (enters validation queue)
- Survey becomes read-only for surveyor

SVR-29: Handle Rework from Validator

As a Surveyor when my survey is sent back for rework **I want to** see the rejection reasons and fix issues. **Acceptance Criteria:**

- Rework status visible on dashboard (Issues category)
- Open survey shows rework banner with:
 - Rejection reason code (e.g., MISSING_PHOTOS, POOR_PHOTO_QUALITY)
 - Free-text remarks from validator
- Survey re-opens for editing
- Status = `rework` (behaves like `on_going` for section visibility)
- After fixing, surveyor can re-submit → status back to `validation_f49`

SVR-30: View Completed Survey (Read-Only)

As a Surveyor I want to review my completed/approved surveys **so that** I can reference past work. **Acceptance Criteria:**

- Completed surveys accessible from History screen
- All sections visible but read-only
- No edit controls rendered

Epic: Surveyor Navigation

SVR-31: Map View of Assigned Surveys

As a Surveyor I want to see my surveys on a map **so that** I can plan my route. **Acceptance Criteria:**

- Leaflet map with survey location markers
- Markers color-coded by status
- Click marker → opens survey detail/card
- Filter by status
- 450m radius nearby building search (Lambert 72 coordinates)

SVR-32: Survey History Timeline

As a Surveyor I want to see a timeline of all my completed surveys. **Acceptance Criteria:**

- Chronological list of completed surveys
- Shows completion date, tsq_id, address
- Click to open read-only view

SVR-33: Surveyor Profile

As a Surveyor I want to view/edit my profile. **Acceptance Criteria:**

- Personal details display
- Regional assignment info
- Statistics (total completed, average time, etc.)

Epic: Signature

SVR-34: Dual-Party Signature

As a Surveyor I want the owner and I to sign the survey statement **so that** there's formal agreement. **Acceptance Criteria:**

- Statement/agreement section at end of survey
- Two signature pads: owner + surveyor
- Timestamp recorded for each signature
- "Refuse to sign" option available
- Visible in later statuses (post on_going)

3. Validator Flows

Epic: Validator Dashboard

VAL-01: View Validation Queue with KPIs

As a Validator I want to see my queue and metrics **so that** I know my review workload. **Acceptance Criteria:**

- KPI cards: Queue Depth (validation_f49 count), Approved (completed + sent + validation_client), Rejected (rework + rejected), Approval Rate %
- KPIs update on each action

VAL-02: Filter Queue by Status

As a Validator I want to filter the queue **so that** I can focus on specific items. **Acceptance Criteria:**

- Filters: All, Pending Review (validation_f49), Approved (completed, sent, validation_client), Rejected (rework, rejected)
- Project filter dropdown
- Active filter highlighted

VAL-03: Toggle Queue View (List / Table / Map)

As a Validator I want to switch between list, table, and map views. **Acceptance Criteria:**

- o List: card-based view
- o Table: sortable columns (tsg_id, address, status, date)
- o Map: Leaflet map with markers
- o View preference persisted

Epic: Validation Workspace

VAL-04: Open Validation Workspace

As a Validator I want to open a survey for review **so that** I can assess quality. **Acceptance Criteria:**

- o Click survey from queue → workspace opens
- o Full survey displayed in read-only mode
- o Previous/Next buttons for queue traversal
- o Queue position indicator (e.g., "3 of 15")

VAL-05: View Pre-Validation Flags

As a Validator I want to see automatic quality flags **so that** I know what to focus on. **Acceptance Criteria:**

- o Bottom panel lists all missing required fields per section
- o Alerts if owner signature is missing
- o Shows rework history if previously rejected
- o Color-coded severity (error = red, warning = amber)

VAL-06: Approve Survey

As a Validator I want to approve a survey **so that** it moves to client review. **Acceptance Criteria:**

- o "Approve" button (green) in action bar
- o Confirmation dialog
- o Status transition: `validation_f49` → `validation_client`
- o Success banner shown
- o Auto-advance to next survey in queue (or return to queue if last)

VAL-07: Reject Survey (Send Back for Rework)

As a Validator I want to reject a survey with reasons **so that** the surveyor knows what to fix. **Acceptance Criteria:**

- o "Reject" button (red) in action bar
- o Rejection dialog opens with:
 - Reason code dropdown (required): MISSING_PHOTOS, MISSING_SIGNATURE, INCORRECT_UNIT_COUNT, INCORRECT_FLOOR_COUNT, MISSING_QUANTITIES, POOR_PHOTO_QUALITY, INCOMPLETE_SECTIONS, DATA_INCONSISTENCY, OTHER
 - Free-text notes field for detailed remarks
- o On submit: status → `rework`, rework_remarks saved
- o Surveyor sees rejection reason + notes
- o Survey returns to queue as rejected

VAL-08: Navigate Queue (Previous/Next)

As a Validator I want to quickly move between surveys in my queue. **Acceptance Criteria:**

- o Previous/Next buttons in workspace header
- o Queue position updates

- Keyboard shortcuts (← →) for power users
- Loops back to start/end of queue

4. PM/Admin Flows

Epic: Executive Dashboard

ADM-01: View Executive Dashboard

As a PM/Admin I want to see system-wide metrics **so that** I have full operational visibility. **Acceptance Criteria:**

- KPI cards: Total Addresses, Completed (count + %), In Progress (%), Pending (%), In Review (%), Issues (%)
- Status distribution bar chart (all 15 statuses)
- Weekly completion trend line chart (completed vs started per week)
- Funnel/Pie chart breakdown
- Quick-link: "Create Survey" button

Epic: Survey Creation

ADM-02: Create Survey Manually

As a PM/Admin I want to manually create a new survey **so that** I can add buildings not in bulk import. **Acceptance Criteria:**

- Form fields: TSG ID (unique), Street, Number, Postal Code, City, Building Type dropdown (single_family, terraced, semi_detached, apartment), Priority checkbox, Assigned Surveyor dropdown
- On submit: creates survey with status = `to_do`
- Auto-generates building_id = `BLD-{tsg_id}`
- Pre-fills empty structures for all 22 form sections
- Validation: TSG ID must be unique
- Returns to Executive Dashboard on success

Epic: Project Management

ADM-03: View Project Dashboard

As a PM/Admin I want to see project-level survey groupings and KPIs. **Acceptance Criteria:**

- List of projects
- Per-project: total surveys, status breakdown, progress %
- Click project → filtered survey list

Epic: Performance Tracking

ADM-04: View Surveyor Performance

As a PM/Admin I want to track surveyor performance **so that** I can manage team productivity. **Acceptance Criteria:**

- Per-surveyor: surveys completed, average time per survey, photo quality metrics
- Regional performance breakdown
- Sortable/filterable table
- Export to CSV

ADM-05: View Subcontractor Performance

As a PM/Admin I want to track subcontractor build performance. **Acceptance Criteria:**

- Per-subco: build progress, dispute frequency, phase completion rates
- Sortable table
- Export to CSV

Epic: Map & Table View

ADM-06: GIS Map + Data Table View

As a PM/Admin I want to see all surveys on a map and in a table **so that** I can manage geographically. **Acceptance Criteria:**

- Leaflet map with all survey markers (color-coded by status)
- Table view alongside/below map
- Filter by status, project, surveyor, city
- Click marker or row → opens survey detail
- Search by tsg_id or address

Epic: Approval Flow

ADM-07: View 9-Step Approval Pipeline

As a PM/Admin I want to track each survey's journey through the full approval pipeline. **Acceptance Criteria:**

- Left panel: list of all surveys in approval pipeline
- Center panel: selected survey's 9-step journey:
 - Survey Completed
 - SSV Created
 - TSA Created
 - Engineering Inputs
 - Ready for Construction
 - Build Status
 - Build Approved (branches: Approved / Dispute)
 - Finances Generated
 - Build Complete
- Each step shows: status (pending/current/completed), actor, timestamp, action link
- Filter: by step, by status (validation_f49, validation_client, completed, sent)

ADM-08: Generate & Send Client PDF (SSV)

As a PM/Admin I want to generate the SSV PDF and send it to the client **so that** they can review/approve. **Acceptance Criteria:**

- Available when survey status = validation_f49 or higher
- "Generate PDF" button creates SSV document
- "Send to Client" action with email/transmission
- Status updates to reflect PDF sent
- Audit log: who sent, when, to whom

ADM-09: Generate & Send Syndic/Owner PDF (TSA)

As a PM/Admin I want to generate and send the syndic/owner PDF **so that** they can sign off. **Acceptance Criteria:**

- **Hard Gate:** Only available after Client Approved (status = validation_client or higher)

- o "Generate Syndic PDF" creates TSA document
- o "Send to Syndic/Owner" action
- o Tracks transmission in audit log

ADM-10: Record Client Approval

As a PM/Admin I want to record client approval **so that** the workflow advances. **Acceptance Criteria:**

- o "Mark Client Approved" action available on validation_client surveys
- o Transitions status appropriately
- o Enables Syndic PDF generation (unlocks gate)

ADM-11: Record Syndic/Owner Approval

As a PM/Admin I want to record syndic/owner approval **so that** engineering can begin. **Acceptance Criteria:**

- o "Mark Syndic Approved" action
- o **Unlocks gate:** Engineering inputs now allowed
- o Status update reflected in pipeline

Epic: Engineering Gate

ADM-12: View Engineering Gate Dashboard

As a PM/Admin I want to see which surveys need engineering inputs **so that** I can prepare them for construction.

Acceptance Criteria:

- o List of surveys eligible for engineering (status = completed, sent, validation_client)
- o Shows gate completion status per survey
- o Filter by completion state

ADM-13: Fill Engineering Gate — Build Type Selection

As a PM/Admin I want to select the build type **so that** the correct construction plan applies. **Acceptance Criteria:**

- o Dropdown: SDU Standard, SDU Complex, Small MDU (9-16 units), Large MDU (17-48 units), Extra Large MDU (49+ units)
- o Selection saved immediately
- o Required for gate completion

ADM-14: Fill Engineering Gate — DB7 & Subduct Color

As a PM/Admin I want to configure DB7 settings and subduct color. **Acceptance Criteria:**

- o DB7 option dropdown
- o Subduct color selection
- o Required for gate completion

ADM-15: Upload Splicing Plans

As a PM/Admin I want to upload splicing plan files **so that** the subcontractor has construction documents.

Acceptance Criteria:

- o 5 upload slots:
- o PDP – Primary Distribution Point
- o DP – Distribution Point
- o POC – Point of Connection
- o BUDI – Building Distribution
- o Floorboxes
- o File upload per slot (PDF, images)

- o At least one file required for gate completion

ADM-16: Upload Blowing/Jetting Plan (Optional)

As a PM/Admin I want to optionally upload a blowing plan. **Acceptance Criteria:**

- o Single file upload slot
- o Optional (not required for gate completion)

ADM-17: Fill Build Type Label & Address

As a PM/Admin I want to add label and address metadata for the build. **Acceptance Criteria:**

- o Label text field (description)
- o Address text field
- o Required for gate completion

ADM-18: Complete Engineering Gate & Assign Subcontractor

As a PM/Admin I want to finalize engineering inputs and assign a subcontractor. **Acceptance Criteria:**

- o Gate completion validation:
 - Build type selected ✓
 - DB7 + subduct color set ✓
 - At least one splicing file uploaded ✓
 - Label + address filled ✓
- o "Assign Subcontractor" button enabled only when gate complete
- o Assignment modal: select subcontractor from dropdown, confirm
- o Updates survey.assigned_subcontractor
- o Status → ready for build phase

Epic: Subcontractor Monitoring

ADM-19: View Subco Monitor Dashboard

As a PM/Admin I want to see all subcontractor assignments and their progress. **Acceptance Criteria:**

- o List of all assignments as cards
- o Each card: tsg_id, address, subcontractor name, current phase (pre_build/during_build/post_build), progress % bar, status badge (pending_acceptance, accepted, in_progress, disputed, completed)
- o Filter by subcontractor, phase, status
- o Click card → admin view of subco's work

ADM-20: View Subcontractor Execution Detail (Admin View)

As a PM/Admin I want to inspect a subcontractor's uploaded work **so that** I can verify quality. **Acceptance Criteria:**

- o View uploaded photos/forms per phase (pre, during, post)
- o View build status updates
- o View meetstaar if uploaded
- o "Raise Dispute" action available

Epic: Dispute Management

ADM-21: View Active Disputes

As a PM/Admin I want to see all open disputes **so that** I can manage build quality issues. **Acceptance Criteria:**

- List: Dispute ID, tsg_id, subcontractor name, status (open/resolved), raised by, date
- Filter: open, resolved, all
- Click to view dispute detail

ADM-22: Raise New Dispute

As a PM/Admin I want to raise a dispute against a subcontractor's work. **Acceptance Criteria:**

- Select survey (dropdown of surveys with assigned subcos only)
- Comment field (issue description)
- Instructions field (what subco must fix)
- On submit: dispute created with status = open
- Dispute flagged on subco's assignment

ADM-23: Resolve Dispute

As a PM/Admin I want to resolve a dispute after the subcontractor addresses the issue. **Acceptance Criteria:**

- View subco's response/re-uploaded evidence
- "Resolve" button → status = resolved
- Resolution timestamp + admin notes logged

Epic: Financials

ADM-24: Review Meetstaat

As a PM/Admin I want to review meetstaat documents **so that** costs are reconciled. **Acceptance Criteria:**

- List of uploaded meetstaat files per survey/subco
- View/download meetstaat PDF
- Mark as reviewed/approved
- Payment status tracking per subcontractor

Epic: Admin — Import & Grouping

ADM-25: Bulk Import Addresses (CSV)

As a PM/Admin I want to bulk import building addresses **so that** surveys are created at scale. **Acceptance Criteria:**

- CSV upload with required columns: tsg_id, street, number, postal_code, city
- Optional columns: building_type, priority, assigned_surveyor
- Validation: duplicate tsg_id check, required fields
- Preview imported rows before confirmation
- Surveys created with status = `to_do`
- Import result summary: created, skipped (duplicates), errors

ADM-26: MDU Grouping — Link Buildings

As a PM/Admin I want to group multiple buildings into one survey **so that** nearby MDUs share a single assessment.

Acceptance Criteria:

- Select main building → ID suffix = {baseID}-A
- Add child buildings → IDs assigned -B, -C, etc.
- Nearby search: 450m radius using Lambert 72 coordinates
- Child building URLs redirect to main survey (cannot edit from child URL)
- Status changes on main propagate to all children

ADM-27: MDU Grouping — Unlink Buildings

As a PM/Admin I want to remove a building from a group. **Acceptance Criteria:**

- Select grouped building → "Unlink" action
- Child gets its own independent survey
- ID suffix removed or reassigned

ADM-28: RBAC — Manage User Roles

As a PM/Admin I want to assign and manage user roles **so that** access is controlled. **Acceptance Criteria:**

- User list with current roles
- Assign/change role per user (PM, Surveyor, Validator, Subcontractor)
- Role change effective immediately
- Audit log of role changes

5. Subcontractor Flows

Epic: Subcontractor Dashboard

SUB-01: View Assignment List

As a Subcontractor I want to see all my assigned builds **so that** I know my workload. **Acceptance Criteria:**

- Card list of all assignments
- Each card: tsg_id, address, city, build type badge, progress % bar, status badge
- Filter by status: pending_acceptance, accepted, in_progress, completed
- Dispute indicator visible if applicable

SUB-02: Accept Assignment Package

As a Subcontractor I want to accept a new build assignment **so that** I can begin work. **Acceptance Criteria:**

- "New Assignment" card for pending_acceptance status
- Message: "Accept this package to begin build execution"
- "Accept Package" button → status transitions to accepted
- "Request Reassignment" button → sends notification to admin

Epic: Build Execution Hub

SUB-03: View Execution Hub Header

As a Subcontractor I want to see build summary info when working on an assignment. **Acceptance Criteria:**

- Header shows: full address, tsg_id, building_id, build type badge, postal code, city
- Construction status dropdown: Start Construction, VC Done, INTRO Done, Dispute/On Hold
- Dispute banner shown if dispute is active (with comment + remediation instructions)

SUB-04: View Available Documents (SSV + Engineering Files)

As a Subcontractor I want to access the survey report and engineering plans **so that** I can execute the build.
Acceptance Criteria:

- SSV document always available (downloadable)
- Engineering files available per type: PDP, DP, POC, BUDI, Floorboxes, Blowing plan

- o File viewer or download for each document

SUB-05: View Key Building Information Panel

As a Subcontractor **I want to** reference building details while working. **Acceptance Criteria:**

- o Collapsible info section with: Address, MRO Zone, POP Zone, Building ID, Building Group ID, Building Type, Number of Units, Layers, Distribution type, Remarks

Epic: Pre-Build Phase

SUB-06: Upload Intro Images

As a Subcontractor **I want to** upload cable entry and site access photos **so that** pre-build evidence is recorded.

Acceptance Criteria:

- o Photo upload (camera on mobile, file select on desktop)
- o Minimum 1 photo required
- o Quadrant selection for location context
- o Optional remarks

SUB-07: Upload Virtual Site Completion (VC) Images

As a Subcontractor **I want to** upload VC images **so that** virtual completion is documented. **Acceptance Criteria:**

- o Minimum 1 photo required
- o Photo upload + metadata
- o Pre-build phase complete when both intro + VC images uploaded and submitted

SUB-08: Submit Pre-Build Phase

As a Subcontractor **I want to** submit pre-build deliverables **so that** I can move to the during-build phase. **Acceptance Criteria:**

- o Validation: intro images ≥ 1 + VC images ≥ 1
- o Submit button → phase transitions to during_build
- o Cannot go back to pre_build after submission

Epic: During-Build Phase

SUB-09: Fill During-Build Form

As a Subcontractor **I want to** record cable routing and riser progress. **Acceptance Criteria:**

- o Cable routing detail fields
- o Riser progress tracking (for MDUs)
- o Floor-by-floor progress (for large MDUs)

SUB-10: Upload During-Build Photos

As a Subcontractor **I want to** upload progress photos **so that** build quality is evidenced. **Acceptance Criteria:**

- o Multiple photo upload slots
- o Evidence of cable placement
- o Safety/compliance documentation photos
- o Required count varies by build type (SDU Standard = 4, Large MDU = 12)

SUB-11: Submit During-Build Phase

As a Subcontractor I want to complete the during-build phase. **Acceptance Criteria:**

- Validation: required form fields + minimum photo count met
- Submit → phase transitions to post_build

Epic: Post-Build Phase

SUB-12: Upload Post-Build Completion Photos

As a Subcontractor I want to upload final completion photos. **Acceptance Criteria:**

- Post-build completion photos (multiple)
- Final testing results upload (for large MDU)
- Handover report upload (for XL MDU)

SUB-13: Upload Meetstaat

As a Subcontractor I want to upload the meetstaat (cost reconciliation) document **so that** financials can be processed. **Acceptance Criteria:**

- Meetstaat file upload (PDF)
- Upload timestamp recorded
- Triggers admin review
- Cannot re-upload once admin has approved (lock after approval)

SUB-14: Submit Post-Build Phase & Complete Build

As a Subcontractor I want to finalize the build execution. **Acceptance Criteria:**

- Validation: all post-build photos + meetstaat uploaded
- Submit → status = completed
- Build becomes read-only

Epic: Subcontractor Dispute Handling

SUB-15: View Dispute & Remediation Instructions

As a Subcontractor when a dispute is raised against me **I want to** see what needs fixing. **Acceptance Criteria:**

- Dispute banner on execution hub
- Shows PM's comment (what's wrong)
- Shows remediation instructions (what to do)
- Assignment status = disputed

SUB-16: Respond to Dispute

As a Subcontractor I want to respond to a dispute with new evidence. **Acceptance Criteria:**

- Re-upload corrected photos
- Add notes explaining changes
- "Mark as Responded" action
- Awaits admin resolution (cannot self-resolve)

6. Cross-Functional Flows

Epic: Auto Status Transitions

SYS-01: Implement Auto Status Transitions

As the System I want to auto-transition survey statuses **so that** the workflow progresses correctly. **Rules:**

1. First save on `to_do` → `on_going`
2. Add visit on `on_going` → `visited`
3. 3+ "No entry" visits → `final_no_entry`
4. Add appointment → `appointment`
5. All status changes propagate to grouped MDU children

SYS-02: Enforce Hard Gate Rules

As the System I want to block actions that violate workflow gates. **Rules:**

1. No Syndic/Owner PDF generation until Client Approved
2. No engineering input access until Syndic/Owner Approved
3. No subcontractor assignment until Engineering Gate complete

Epic: MDU Grouping Behavior

SYS-03: Grouped MDU Redirects

As the System when a user accesses a child MDU URL **then** redirect to the main survey. **Acceptance Criteria:**

- Child building URL → redirect to main building's survey
- Edit only happens on main survey
- All child buildings display main survey data

SYS-04: Status Propagation to Grouped MDUs

As the System when the main survey status changes **then** update all children. **Acceptance Criteria:**

- Any status change on main → same status set on all children
- Works for both manual and auto-transitions

Epic: Offline Capability (Surveyor)

SYS-05: Offline Data Entry

As a Surveyor I want to fill surveys offline **so that** I can work in buildings with poor connectivity. **Acceptance Criteria:**

- Survey data cached locally
- Edits queued when offline
- Auto-sync when connection restored
- Conflict resolution: last-write-wins with user notification

SYS-06: Offline Photo Capture

As a Surveyor I want to take photos offline **so that** I can work without interruption. **Acceptance Criteria:**

- Photos stored locally
- Queued for upload when online

- Upload progress indicator on sync

Epic: Bilingual Support (EN/NL)

SYS-07: Language Toggle

As a User I want to switch between English and Dutch **so that** I can work in my preferred language.

Acceptance Criteria:

- Language toggle in profile/settings
- All UI labels and messages available in EN and NL
- Survey form labels default to Dutch (NL)
- Persisted preference per user

Epic: Responsive / Mobile-First

SYS-08: Mobile Layout — Surveyor

Acceptance Criteria:

- Bottom navigation bar (5 items: Dashboard, Map, History, Profile)
- Bottom nav hides when survey form is open (back button instead)
- Full-width survey sections, vertical scroll
- Camera integration for photo capture

SYS-09: Mobile Layout — Validator

Acceptance Criteria:

- List view dominant (not table)
- Filters collapse to dropdown
- Full-screen workspace for review

SYS-10: Mobile Layout — Admin

Acceptance Criteria:

- Bottom navigation with "More" dropdown for overflow tabs
- Charts stack vertically
- Tables become scrollable cards

SYS-11: Mobile Layout — Subcontractor

Acceptance Criteria:

- Vertical card stack for assignments
- Full-screen execution hub with document viewer

Appendix: Permission Matrix

| Permission | Surveyor | Validator | Admin | Subco |

|---|:---|:---|:---|:---|

| survey.view_assigned | ✓ | | | |

| survey.view_all | | ✓ | ✓ | |

survey.edit	✓		✓	
survey.submit	✓			
survey.validate		✓	✓	
survey.approve		✓	✓	
survey.reject		✓	✓	
visit.add/delete	✓			
appointment.add/delete	✓			
photo.upload	✓			✓
photo.delete	✓			
agreement.sign	✓			✓
validation.add_notes		✓		
workflow.manage			✓	
pdf.generate/send			✓	
engineering.upload			✓	
engineering.define_build_type			✓	
subcontractor.assign			✓	
subcontractor.dispute			✓	
assignment.view_own				✓
assignment.accept				✓
build.upload_photos				✓
build.fill_forms				✓
build.submit_phase				✓
meetstaat.upload				✓
dispute.view/respond			✓	✓

Ticket ID Summary

Prefix	Role	Count
AUTH	All	3
SVR	Surveyor	34
VAL	Validator	8
ADM	PM/Admin	28
SUB	Subcontractor	16
SYS	Cross-functional	11
Total		**100**